- Solver : reactingFoam
https://gitlab.com/openfoam/core/openfoam/-/blob/master/applications/solvers/combustion/reactingFoam/reactingFoam.C

- Navier Stokes Compressible
https://gitlab.com/openfoam/core/openfoam/-/blob/master/applications/solvers/combustion/reactingFoam/UEqn.H?ref_type=heads

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U}\mathbf{U}) = -\nabla p + \nabla \cdot \boldsymbol{\tau}_{\text{eff}} + \rho \mathbf{g}$$

- Transport
https://gitlab.com/openfoam/core/openfoam/-/blob/master/ap[illegible]qn.H?ref_type=heads

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho \mathbf{U} Y_i) = \nabla \cdot (\mu_{\text{eff}} \nabla Y_i) + \dot{\omega}_i, \quad \text{with} \quad \sum_i Y_i = 1$$

- Energie
https://gitlab.com/openfoam/core/openfoam/-/blob/master/applications/solvers/combustion/reactingFoam/EEqn.H?ref_type=heads

$$\frac{\partial(\rho h_s)}{\partial t} + \nabla \cdot (\rho \mathbf{U} h_s) + \frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{U} K) - \frac{\partial p}{\partial t} = \nabla \cdot (\alpha_{\text{eff}} \nabla h_s) + \dot{Q}_{\text{chem}}$$

$$K = |\mathbf{U}|^2/2$$

$$\dot{Q}_{\text{chem}} = -\sum_i h^{\circ}_{f,i}\, \dot{\omega}_i.$$

- Fermeture combustion (PaSR : Partially Stirred Reactor)
https://gitlab.com/openfoam/core/openfoam/-/blob/master/src/combustionModels/PaSR/PaSR.C

$$\dot{\omega}_i = \kappa\, \dot{\omega}_i^{\text{chem}}, \qquad \kappa = \frac{\tau_c}{\tau_c + \tau_{\text{mix}}}, \qquad \tau_{\text{mix}} = C_{\text{mix}} \sqrt{\frac{\nu_{\text{eff}}}{\varepsilon}}$$

- Equation d’état

https://gitlab.com/openfoam/core/openfoam/-/blob/master/src/thermophysicalModels/specie/equationOfState/perfectGas/perfectGasI.H

$$p = \rho \frac{R}{W} T, \qquad \psi \equiv \frac{1}{RT/W} \implies \rho = \psi p$$